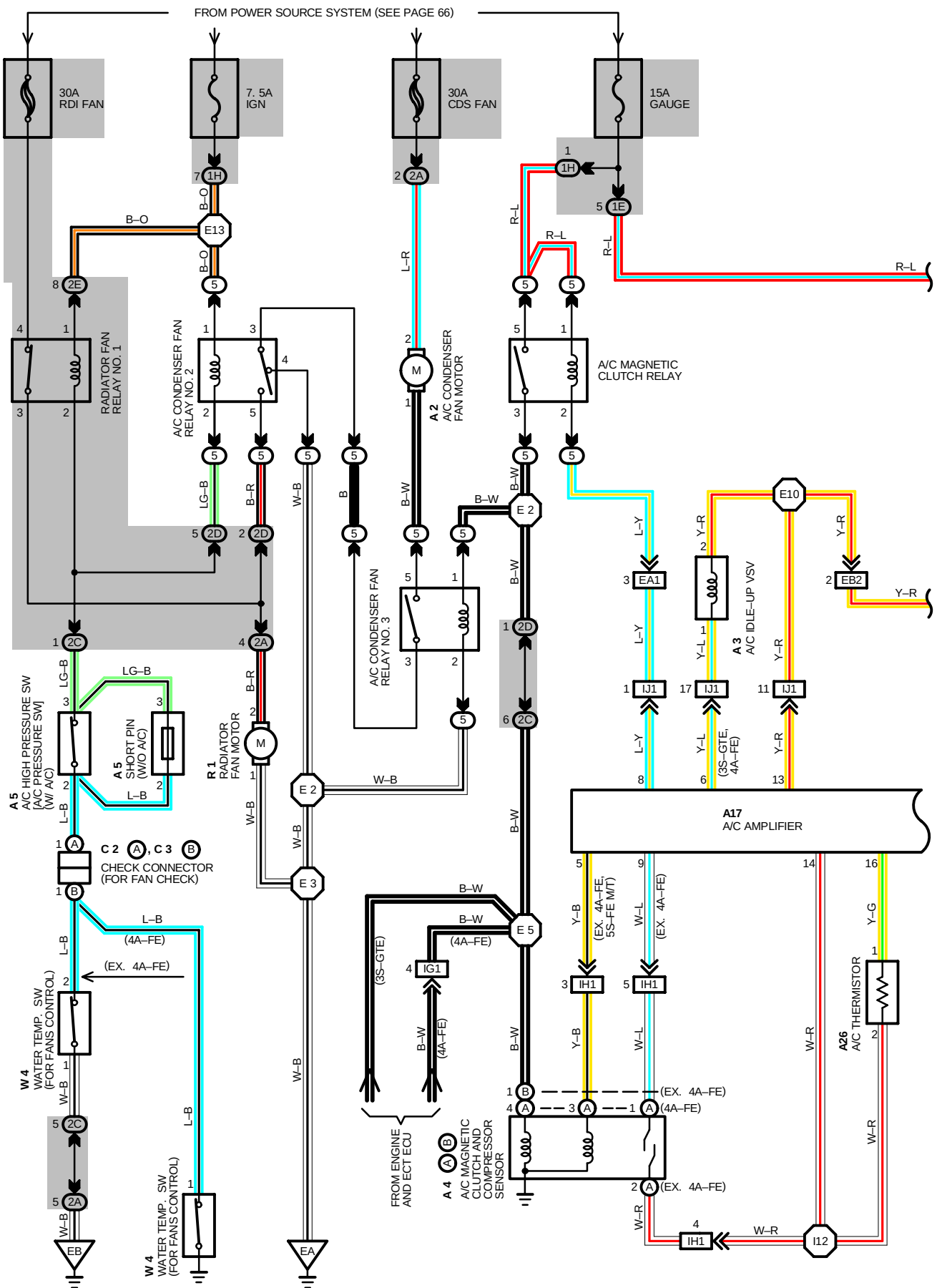
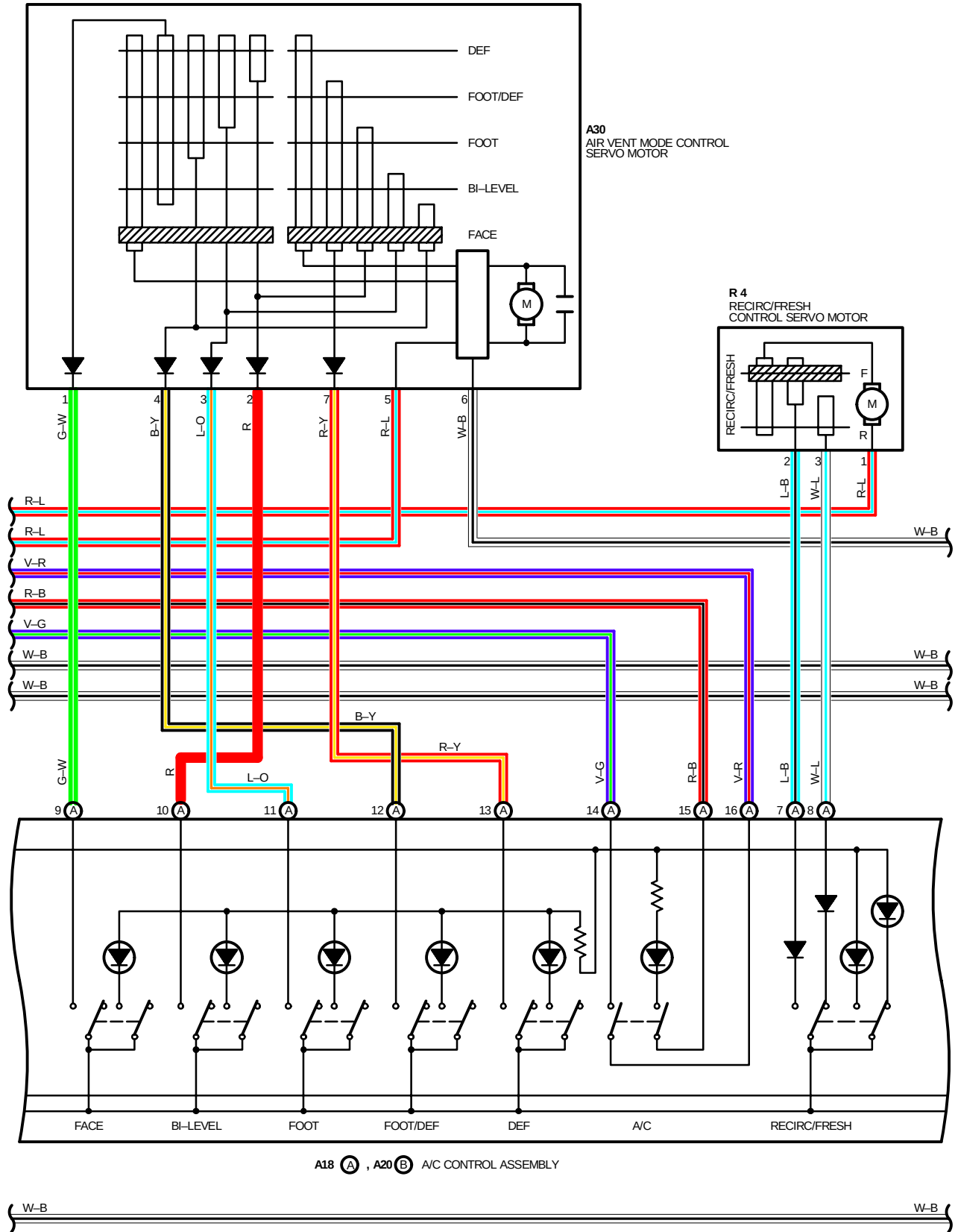


# RADIATOR FAN AND AIR CONDITIONER (MANUAL AIR CONDITIONER)





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## SYSTEM OUTLINE

### 1. COOLING FAN OPERATION

WHEN THE IGNITION SW IS TURNED ON, THE CURRENT FROM IGN FUSE FLOWS TO **TERMINAL 1** OF RADIATOR FAN RELAY NO. 1 → **TERMINAL 2** → **TERMINAL 3** OF THE A/C PRESSURE SW → **TERMINAL 2** → **TERMINAL 1** (4S-FE) OR **TERMINAL 2** (EX. 4A-FE) OF THE WATER TEMP. SW → **TERMINAL 1** (EX. 4A-FE) → **GROUND**, FROM **TERMINAL 1** OF A/C FAN RELAY NO. 2 → **TERMINAL 2** → **TERMINAL 3** OF A/C PRESSURE SW → **TERMINAL 2** → **TERMINAL 1** (4A-FE) OR **TERMINAL 2** (EX. 4A-FE) OF THE WATER TEMP. SW → **TERMINAL 1** (EX. 4A-FE) → **GROUND**, CAUSING RELAY NO. 1 AND RELAY NO. 2 OF EACH FAN TO TURN ON.

#### \* OPERATION AT LOW SPEED

WHEN THE A/C SW (A/C CONTROL ASSEMBLY) IS TURNED ON AND THE AIR CONDITIONER OPERATES, THE CURRENT FLOWS FROM GAUGE FUSE FLOWS TO **TERMINAL 1** OF A/C MAGNETIC CLUTCH RELAY → **TERMINAL 2** → **TERMINAL 8** OF A/C AMPLIFIER CAUSING A/C MAGNETIC CLUTCH RELAY TO TURN ON.

AT THAT TIME, THE CURRENT FROM GAUGE FUSE FLOWS TO **TERMINAL 5** OF A/C MAGNETIC CLUTCH RELAY → **TERMINAL 3** → **TERMINAL 4** (5S-FE), **1** (4A-FE) OF A/C MAGNETIC CLUTCH → **GROUND**, AND FROM **TERMINAL 3** OF A/C MAGNETIC CLUTCH RELAY → **TERMINAL 2** OF A/C FAN RELAY NO. 3 → **TERMINAL 2** → **GROUND**.

AS A RESULT, A/C MAGNETIC CLUTCH AND A/C FAN RELAY NO. 3 TURNS ON AND THE CURRENT FLOWS FROM CDS FAN FUSE → **TERMINAL 2** OF A/C CONDENSER FAN MOTOR → **TERMINAL 1** → **TERMINAL 5** OF A/C FAN RELAY NO. 3 → **TERMINAL 3** → **TERMINAL 3** OF A/C FAN RELAY NO. 2 → **TERMINAL 5** → **TERMINAL 2** OF A/C CONDENSER FAN MOTOR → **TERMINAL 1** → **GROUND**, FLOWING TO EACH FAN MOTOR IN SERIES, CAUSING THE COOLING FAN TO ROTATE AT LOW SPEED.

#### \* OPERATION AT HIGH SPEED

DURING A/C OPERATION, WHEN THE PRESSURE OF A/C COMPRESSOR BECOMES HIGHER THAN NORMAL PRESSURE (MORE THAN 14.3 KG/CM<sup>2</sup> 1401 KPA, 203 PSI), THE A/C PRESSURE SW TURNS OFF.

AS A RESULT, FAN RELAY NO. 1 AND NO. 2 TURNS OFF AND THE CURRENT FLOWS FROM RDI FAN FUSE → **TERMINAL 4** OF RADIATOR FAN RELAY NO. 1 → **TERMINAL 3** → **TERMINAL 2** OF RADIATOR FAN MOTOR → **TERMINAL 1** → **GROUND**, AND FROM CDS FAN FUSE → **TERMINAL 2** OF A/C CONDENSER FAN MOTOR → **TERMINAL 1** → **TERMINAL 5** OF A/C FAN RELAY NO. 3 → **TERMINAL 3** → **TERMINAL 3** OF A/C FAN RELAY NO. 2 → **TERMINAL 4** → **GROUND**, FLOWING TO EACH FAN MOTOR IN PARALLEL CAUSING THE COOLING FAN TO ROTATE AT HIGH SPEED. WHEN THE ENGINE COOLANT TEMPERATURE BECOMES MORE THAN ABOUT 90°C (194°F), THE WATER TEMP. SW TURNS OFF AND THE SAME OPERATION AS ABOVE IS PERFORMED.

### 2. HEATER BLOWER MOTOR OPERATION (DIAL TYPE BLOWER (W/O AUTO A/C))

CURRENT IS APPLIED AT ALL TIMES THROUGH THE HEATER FUSE TO **TERMINAL 5** OF HEATER RELAY. WHEN THE IGNITION SW IS TURNED ON, CURRENT FLOWS THROUGH GAUGE FUSE TO **TERMINAL 3** OF HEATER RELAY → **TERMINAL 1** → **TERMINAL (B) 2** AND **TERMINAL (B) 7** OF A/C CONTROL ASSEMBLY.

#### \* LOW SPEED OPERATION

WHEN THE BLOWER SW (A/C CONTROL ASSEMBLY) IS MOVED TO **LO** POSITION. THE CURRENT FLOWS TO **TERMINAL (B) 7** OF THE A/C CONTROL ASSEMBLY → **TERMINAL (B) 3** → **GROUND** AND TURNS THE HEATER RELAY ON.

AS A RESULT, THE CURRENT TO **TERMINAL 5** OF THE HEATER RELAY → **TERMINAL 4** → **TERMINAL 2** OF BLOWER MOTOR → **TERMINAL 1** → **TERMINAL 1** OF BLOWER RESISTOR → **TERMINAL 4** → **GROUND**, CAUSING THE BLOWER MOTOR TO ROTATE AT LOW SPEED.

#### \* HIGH SPEED OPERATION

WHEN THE BLOWER SW (A/C CONTROL ASSEMBLY) IS MOVED TO **HI** POSITION. THE CURRENT TO **TERMINAL (B) 2** OF A/C CONTROL ASSEMBLY → **TERMINAL (B) 3** → **GROUND** AND TURNS THE HEATER RELAY ON.

AS A RESULT, THE CURRENT TO **TERMINAL 5** OF HEATER RELAY → **TERMINAL 4** → **TERMINAL 2** OF BLOWER MOTOR → **TERMINAL 1** → **TERMINAL (B) 5** OF A/C CONTROL ASSEMBLY → **TERMINAL (B) 3** → **GROUND**, AND CAUSES THE BLOWER MOTOR TO ROTATE AT HIGH SPEED.

#### \* MEDIUM SPEED OPERATION (OPERATION AT MANUAL M1, M2)

WHEN THE BLOWER SW (A/C CONTROL ASSEMBLY) IS MOVED TO **M1** POSITION. THE CURRENT FLOWING TO **TERMINAL 3** OF HEATER RELAY FLOWS TO **TERMINAL 1** → **TERMINAL (B) 2** OF A/C CONTROL ASSEMBLY → **TERMINAL (B) 3** → **GROUND** AND TURNS THE HEATER RELAY ON.

AS A RESULT, CURRENT FROM THE HEATER FUSE TO **TERMINAL 5** OF HEATER RELAY FLOW TO **TERMINAL 4** → **TERMINAL 2** OF BLOWER MOTOR → **TERMINAL 1** OF BLOWER RESISTOR → **TERMINAL 3** → **TERMINAL (B) 4** OF A/C CONTROL ASSEMBLY → **TERMINAL (B) 3** → **GROUND**, CAUSING THE BLOWER MOTOR TO ROTATE AT MEDIUM LOW SPEED.

WHEN THE BLOWER SW (A/C CONTROL ASSEMBLY) IS MOVED TO **M2** POSITION. THE CURRENT TO **TERMINAL 5** OF HEATER RELAY FLOWS TO **TERMINAL 4** → **TERMINAL 2** OF BLOWER MOTOR → **TERMINAL 1** → **TERMINAL 1** OF BLOWER RESISTOR → **TERMINAL 2** → **TERMINAL (B) 6** OF A/C CONTROL ASSEMBLY → **TERMINAL (B) 3** → **GROUND**.

AS THE CURRENT FLOW FROM BLOWER MOTOR TO GROUND IS GREATER THAN FOR M1. THE BLOWER MOTOR ROTATES AT HIGH SPEED.

### 3. OPERATION OF RECIRC/FRESH CONTROL SERVO MOTOR

(SWITCHING FROM FRESH TO RECIRC)

WITH IGNITION SW TURNED ON, THE CURRENT FLOWS FROM GAUGE FUSE TO **TERMINAL 1** OF AIR INLET CONTROL SERVO MOTOR. WHEN THE RECIRC/FRESH SW IS SWITCHED TO THE RECIRC SIDE, THE CURRENT FLOWS FROM **TERMINAL 1** OF AIR INLET CONTROL SERVO MOTOR → **TERMINAL 2** → **TERMINAL (A) 7** OF RECIRC/FRESH CONTROL SERVO MOTOR → **TERMINAL (A) 2** → **GROUND**, THE MOTOR ROTATES AND THE DAMPER MOVES TO THE RECIRC SIDE.

WHEN IT IS IN THE **RECIRC** POSITION, THE CURRENT IS CUT INSIDE THE SERVO MOTOR AND THE DAMPER STOPS AT THAT POSITION.

(SWITCHING FROM RECIRC TO FRESH)

WITH IGNITION SW ON, WHEN THE RECIRC/FRESH SW IS SWITCHED TO THE FRESH SIDE, THE CURRENT FLOWS FROM **TERMINAL 1** OF AIR INLET CONTROL SERVO MOTOR → **TERMINAL 3** → **TERMINAL (A) 8** OF A/C CONTROL ASSEMBLY → **TERMINAL (A) 2** → **GROUND**, THE MOTOR ROTATES AND THE DAMPER MOVES TO THE FRESH SIDE. WHEN IT IS IN THE **FRESH** POSITION, THE CURRENT IS CUT INSIDE THE SERVO MOTOR AND THE DAMPER STOPS AT THAT POSITION.

### 4. OPERATION OF AIR VENT MODE CONTROL SERVO MOTOR

WITH IGNITION SW TURNED ON, THE CURRENT FLOWS FROM GAUGE FUSE TO **TERMINAL 5** OF AIR VENT MODE CONTROL SERVO MOTOR → **TERMINAL 6** → **GROUND**, AND THE DAMPER MOVES TO THE POSITION OF THE MODE SELECTION SW OF THE CONTROL ASSEMBLY SWITCH IS ON.

WHEN THE MODE SELECTION SW OF A/C CONTROL ASSEMBLY IS MOVED TO **DEF** POSITION FROM THE DAMPER IN THE **FACE** POSITION, THE CURRENT FLOWS FROM **TERMINAL 7** OF AIR VENT MODE CONTROL SERVO MOTOR TO **TERMINAL (A) 13** OF A/C CONTROL ASSEMBLY → **TERMINAL (A) 2** → **GROUND**.

AS A RESULT, TO SERVO MOTOR OPERATES UNTIL THE DAMPER REACHES **DEF** POSITION.

WHEN THIS OCCURS THE CURRENT TO THE A/C CONTROL ASSEMBLY IS SHUT OFF AND ROTATION OF THE MOTOR STOPS. SWITCHING TO OTHER MODES IS CONTROLLED BY THE SERVO MOTOR ACCORDING THE FLOWING CURRENT:

1. **FOOT/DEF** POSITION: THE CURRENT FLOWS FROM **TERMINAL 4** OF SERVO MOTOR TO **TERMINAL (A) 12** OF A/C CONTROL ASSEMBLY.
2. **FOOT** POSITION: THE CURRENT FLOWS FROM **TERMINAL 3** OF SERVO MOTOR TO **TERMINAL (A) 11** OF A/C CONTROL ASSEMBLY.
3. **BI-LEVEL** POSITION: THE CURRENT FLOWS FROM **TERMINAL 2** OF SERVO MOTOR TO **TERMINAL (A) 10** OF A/C CONTROL ASSEMBLY.

### 5. AIR CONDITIONER OPERATION

THE A/C AMPLIFIER RECEIVES VARIOUS SIGNALS, THE ENGINE RPM SIGNAL FROM THE IGNITER, OUTLET TEMPERATURE SIGNAL FROM THE A/C THERMISTOR, COOLANT TEMPERATURE FROM THE WATER TEMP. SENSOR, AND THE LOCK SIGNAL FROM THE A/C COMPRESSOR, ETC.

WHEN THE ENGINE IS STARTED AND THE A/C SW (A/C CONTROL ASSEMBLY) IS TURNED ON, THE CURRENT FLOWS FROM A/C FUSE TO **TERMINAL 13** OF A/C AMPLIFIER.

AS A RESULT, THE CURRENT FLOWING FROM **TERMINAL 2** OF A/C MAGNETIC CLUTCH RELAY TO **TERMINAL 8** OF A/C AMPLIFIER FLOWS FROM **TERMINAL 15** OF A/C AMPLIFIER TO **GROUND** AND TURNS IN THE MAGNETIC CLUTCH RELAY.

BECAUSE THE MAGNETIC CLUTCH IS ON, THE A/C COMPRESSOR OPERATES, CAUSING THE CURRENT FLOWING FROM A/C IDLE-UP VSV TO **TERMINAL 6** OF A/C AMPLIFIER TO FROM TO **TERMINAL 15** OF A/C AMPLIFIER → **GROUND**, AND TURNS ON THE VSV TO AVOID LOWERING THE ENGINE RPM DURING AIR CONDITIONER OPERATION.

WHEN ANY OF THE FOLLOWING SIGNALS ARE INPUT TO THE A/C AMPLIFIER, THE AMPLIFIER OPERATES TO TURN OFF THE AIR CONDITIONER:

- \* ENGINE HIGH RPM SIGNAL
- \* COOLANT TEMP. SIGNAL IS HIGH
- \* A SIGNAL THAT THE TEMPERATURE AT THE AIR OUTLET IS LOW
- \* A SIGNAL THAT THERE IS A LARGE DIFFERENCE BETWEEN ENGINE RPM AND COMPRESSOR RPM
- \* A SIGNAL THAT THE REFRIGERANT PRESSURE IS ABNORMALLY HIGH OR LOW

# RADIATOR FAN AND AIR CONDITIONER (MANUAL AIR CONDITIONER)

## SERVICE HINTS

### A 4 (A)(B) A/C MAGNETIC CLUTCH

(A)4, (B)1—GROUND : APPROX. 3.7 Ω

### A 5 A/C PRESSURE SW

3-2 : OPEN ABOVE APPROX. 13.5 KG/CM<sup>2</sup> (192 PSI, 1323 KPA)

CLOSED BELOW APPROX. 10 KG/CM<sup>2</sup> (142 PSI, 980 KPA)

1-4 : OPEN WITH PRESSURE LESS THAN 2.1 KG/CM<sup>2</sup> (30 PSI, 206 KPA) OR ABOVE 27 KG/CM<sup>2</sup> (384 PSI, 2648 KPA)

### A17 A/C AMPLIFIER

8-15 : CONTINUITY WITH A/C SW (A/C CONTROL ASSEMBLY) ON AND IGNITION SW AT ON POSITION

14-15 : ALWAYS CONTINUITY

14-GROUND: ALWAYS CONTINUITY

15-GROUND: ALWAYS CONTINUITY

13-GROUND: APPROX. 12 VOLTS WITH IGNITION SW ON

### A26 A/C THERMISTOR

1-2 : APPROX. 4852 Ω AT 0°C (32°F)

APPROX. 2341 Ω AT 15°C (39°F)

APPROX. 1500 Ω AT 25°C (77°F)

### B 6 BLOWER RESISTOR

1-3 : APPROX. 0.48 Ω

3-2 : APPROX. 0.94 Ω

2-4 : APPROX. 0.91 Ω

### W 4 WATER TEMP. SW (FOR FANS CONTROL)

1-2 : OPEN ABOVE APPROX. 90°C (194°F)

CLOSED BELOW APPROX. 83°C (181.4°F)



## : PARTS LOCATION

| CODE | SEE PAGE                              | CODE  | SEE PAGE                            | CODE | SEE PAGE                            |
|------|---------------------------------------|-------|-------------------------------------|------|-------------------------------------|
| A 2  | 28 (3S-GTE), 30 (5S-FE), 31 (4A-FE)   | A20 B | 32                                  | J 4  | 33                                  |
| A 3  | 28 (3S-GTE), 30 (5S-FE), 31 (4A-FE)   | A26   | 32                                  | J 5  | 33                                  |
| A 4  | A 28 (3S-GTE), 31 (4A-FE)             | A30   | 32                                  | J 6  | 33                                  |
|      | B 28 (3S-GTE), 30 (5S-FE), 31 (4A-FE) | B 4   | 32                                  | R 1  | 28 (3S-GTE), 30 (5S-FE), 31 (4A-FE) |
| A 5  | 28 (3S-GTE), 30 (5S-FE), 31 (4A-FE)   | B 6   | 32                                  | R 4  | 33                                  |
| A17  | 32                                    | C 2 A | 28 (3S-GTE), 30 (5S-FE), 31 (4A-FE) | W 4  | 28 (3S-GTE), 30 (5S-FE), 31 (4A-FE) |
| A18  | A 32                                  | C 3 B | 28 (3S-GTE), 30 (5S-FE), 31 (4A-FE) |      |                                     |



## : RELAY BLOCKS

| CODE | SEE PAGE | RELAY BLOCKS (RELAY BLOCK LOCATION)        |
|------|----------|--------------------------------------------|
| 4    | 27       | R/B NO. 4 (RIGHT KICK PANEL)               |
| 5    | 27       | R/B NO. 5 (ENGINE COMPARTMENT FRONT RIGHT) |



## : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

| CODE | SEE PAGE | JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)   |
|------|----------|--------------------------------------------------------|
| 1A   | 20       | COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)              |
| 1E   |          |                                                        |
| 1H   | 20       | ENGINE ROOM MAIN WIRE AND J/B NO. 1 (LEFT KICK PANEL)  |
| 2A   | 22 (2WD) | ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY) |
| 2C   | 22 (2WD) | ENGINE WIRE AND J/B NO. 2 (NEAR THE BATTERY)           |
| 2D   | 22 (2WD) | ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY) |
| 2E   |          |                                                        |
| 3A   | 25       | COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)     |



## : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

| CODE | SEE PAGE    | JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)             |
|------|-------------|------------------------------------------------------------------------|
| EA1  | 38 (3S-GTE) | COWL WIRE AND ENGINE ROOM MAIN WIRE (FRONT SIDE OF RIGHT FRONT FENDER) |
|      | 40 (5S-FE)  |                                                                        |
|      | 42 (4A-FE)  |                                                                        |
| EB2  | 38 (3S-GTE) | ENGINE WIRE AND COWL WIRE (REAR SIDE OF RIGHT FRONT FENDER)            |
|      | 40 (5S-FE)  |                                                                        |
|      | 42 (4A-FE)  |                                                                        |
| IG1  | 44          | ENGINE WIRE AND COWL WIRE (UNDER THE ENGINE ECU)                       |
| IH1  | 44          | ENGINE WIRE AND A/C NO. 1 WIRE (BEHIND THE GLOVE BOX)                  |
| IJ1  | 46          | COWL WIRE AND A/C NO. 1 WIRE (BEHIND THE GLOVE BOX)                    |
| IQ2  | 46          | COWL WIRE AND A/C NO. 2 WIRE (BESIDE HEATER UNIT)                      |

 : GROUND POINTS

| CODE | SEE PAGE    | GROUND POINTS LOCATION |
|------|-------------|------------------------|
| EA   | 38 (3S-GTE) | FRONT RIGHT FENDER     |
|      | 40 (5S-FE)  |                        |
|      | 42 (4A-FE)  |                        |
| EB   | 38 (3S-GTE) | FRONT LEFT FENDER      |
|      | 40 (5S-FE)  |                        |
| ID   | 44          | LEFT KICK PANEL        |
| IG   | 44          | R/B NO. 4 SET BOLT     |

 : SPLICE POINTS

| CODE       | SEE PAGE    | WIRE HARNESS WITH SPLICE POINTS | CODE           | SEE PAGE    | WIRE HARNESS WITH SPLICE POINTS |
|------------|-------------|---------------------------------|----------------|-------------|---------------------------------|
| E 2        | 38 (3S-GTE) | ENGINE ROOM MAIN WIRE           | E10            | 40 (5S-FE)  | COWL WIRE                       |
|            | 40 (5S-FE)  |                                 |                | 42 (4A-FE)  |                                 |
|            | 42 (4A-FE)  |                                 | E13            | 38 (3S-GTE) | ENGINE ROOM MAIN WIRE           |
| E 3        | 38 (3S-GTE) |                                 |                | 40 (5S-FE)  |                                 |
|            | 40 (5S-FE)  |                                 |                | 42 (4A-FE)  |                                 |
| 42 (4A-FE) | I 3         |                                 | 46             | COWL WIRE   |                                 |
| E 5        | 38 (3S-GTE) | I 8                             |                |             |                                 |
|            | 40 (5S-FE)  | I 9                             |                |             |                                 |
| 42 (4A-FE) | I12         | 46                              | A/C NO. 1 WIRE |             |                                 |
| E10        | 38 (3S-GTE) | COWL WIRE                       |                |             |                                 |

A 2 BLACK



A 3 BLACK



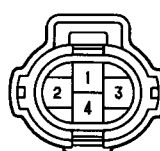
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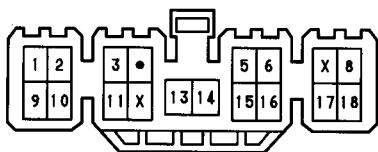
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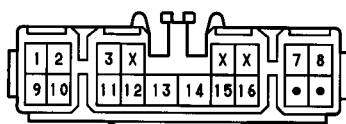
A 5 BLACK



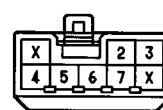
A17



A18 (A)



A20 (B)



A26



A30



B 4 (A) BLACK



B 6 BLACK



C 2 (A) GRAY



C 3 (B) GRAY

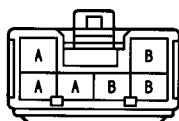


J 4



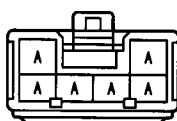
(HINT:SEE PAGE 7)

J 5



(HINT:SEE PAGE 7)

J 6



(HINT:SEE PAGE 7)

R 1



R 4



(EX. 4A-FE) W 4 GRAY



(4A-FE) W 4 GRAY

